

# Mahdi Mohammadigohari

Machine Learning Theory ■ Functional Analysis ■ Kernel and Optimization Geometry

Independent Researcher, Bolzano, Italy

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## Research Profile

I develop mathematical foundations for representation learning and deep models using reproducing-kernel Hilbert spaces, variation spaces, operator theory, empirical-process methods, and variational dynamics. My current research program centers on Brownian kernel ladders: recursive function spaces whose geometry can be studied across representation, statistical complexity, approximation, covariance compression, optimization, and mean-field limits. Complementary work develops Koopman- and composition-operator approaches to multi-task generalization. I am particularly interested in rigorous links between adaptive representations, observable quotient geometry, nonsmooth optimization, and certifiable algorithms.

Recursive RKHSs and variation spaces ■ Statistical learning theory ■ Gradient flows and mean-field limits ■ Operator-theoretic multi-task learning

## Selected Research Contributions

- **Brownian kernel ladder program.** Developed the foundational recursive integral-RKHS construction and extended it to path-atomic variation spaces, common-covariance geometry, simultaneous approximation, finite certification, and current quotient/relaxed dynamics.
- **Representation selection and Brownian heads.** Derived fixed-representation activation-mass characterizations and same-sample selection bounds through Brownian threshold traces for rectangular and rank-deficient ReLU feature extractors.
- **Optimization geometry.** Isolated how equivalent nodal, increment, and spectral realizations of one Brownian RKHS induce identical or genuinely different optimizer trajectories, including exact Brownian-metric preconditioning and basis dependence of Adam.
- **Operator-theoretic generalization.** Developed Koopman/composition-operator bounds for deep multi-output models in vector-valued Sobolev and Brownian RKHS regimes, together with finite-rank operator learning and transfer guarantees.

## Academic and Research Experience

### Independent Researcher in Machine Learning Theory

Jul 2026–Present

*Bolzano, Italy*

Developing the Brownian kernel ladder research program, including observable quotient geometry, relaxed multilayer dynamics, adaptive covariance geometry, representation-selection theory, and optimization-aware finite realizations.

### Doctoral Researcher

Nov 2022–Jun 2026

*Faculty of Engineering, Free University of Bozen–Bolzano, Italy*

Research on statistical learning theory, multi-task deep learning, operator-theoretic generalization, vector-valued RKHSs, and hierarchical Brownian function spaces. Advisor: Prof. Giuseppe Di Fatta.

### Faculty Member / Instructor

2006–2020

*Islamic Azad University, Iran*

Taught undergraduate mathematics and statistics, including Linear Algebra, Real Analysis, Probability, Statistical Inference, Differential Equations, and Numerical Analysis.

## Education

### Ph.D. in Computer Science – Machine Learning Theory

Nov 2022–Sep 2026

*Free University of Bozen–Bolzano, Italy*

Dissertation: *Mathematics of Deep Learning: Generalization in Single-Task and Multi-Task Architectures via Operator-Theoretic and Kernel-Based Methods*. Advisor: Prof. Giuseppe Di Fatta.

### Graduate Studies in Data Analytics (M.S. curriculum)

Sep 2020–May 2022

*Rochester Institute of Technology, RIT Dubai*

GPA: 3.7/4.0

Coursework and Master's Project completed; degree not conferred following a visa interruption. Project: *Energy Consumption Forecasting Using Machine Learning*.

### Ph.D. in Mathematics – Functional Analysis

Sep 2014–Mar 2018

*Islamic Azad University, Mashhad, Iran*

GPA: 18.7/20

Dissertation: *Operator Inequalities and Operators Preserving Orthogonality*. Advisor: Prof. Maryam Amyari.

### M.S. in Pure Mathematics

Sep 2003–Feb 2006

*Islamic Azad University, Mashhad, Iran*

### B.S. in Applied Mathematics

Sep 1999–Sep 2003

*Islamic Azad University, Mashhad, Iran*

## Publications and Manuscripts

### Preprints and submitted manuscripts

- [1] **M. Mohammadigohari**. *Variation Brownian Kernel Ladders*. Submitted manuscript, 2026. [arXiv]
- [2] **M. Mohammadigohari**, G. Di Fatta, G. Nicosia, and P. M. Pardalos. *Brownian Kernel Ladders*. Submitted manuscript, 2026. [arXiv]
- [3] **M. Mohammadigohari**, T. Borsani, and G. Di Fatta. *Operator-Theoretic Generalization Bounds for Multitask Deep Learning*. arXiv preprint, 2026. [arXiv]
- [4] **M. Mohammadigohari**. *Common Covariance Geometry and Certification for Brownian Kernel Ladders*. Submitted manuscript, 2026. [arXiv]
- [5] **M. Mohammadigohari**. *Sample-Weighted End-to-End Trace-Norm Geometry for Multitask Learning*. Submitted manuscript, 2026. [arXiv]
- [6] **M. Mohammadigohari** and N. Mücke. *Brownian Heads for Deep ReLU Representations: Activation Mass and the Cost of Same-Sample Selection*. Submitted manuscript, 2026. [arXiv]
- [7] **M. Mohammadigohari** and G. Camps-Valls. *Optimization Geometry of Equivalent Brownian RKHS Representations*. Submitted manuscript, 2026. [arXiv]

### Peer-reviewed conference proceedings

- [8] **M. Mohammadigohari**, G. Di Fatta, G. Nicosia, and P. M. Pardalos. *On the Koopman-Based Generalization Bounds for Multi-Task Deep Learning*. In Machine Learning, Optimization, and Data Science (LOD 2025), LNCS 16468, Springer, pp. 376–392, 2026. [DOI] [arXiv]
- [9] **M. Mohammadigohari**, G. Di Fatta, G. Nicosia, and P. M. Pardalos. *Operator-Based Generalization Bound for Deep Learning: Insights on Multi-Task Learning*. In Machine Learning, Optimization, and Data Science (LOD 2025), LNCS 16468, Springer, pp. 120–137, 2026. [DOI] [arXiv]

### Journal articles in functional analysis and applied mathematics

- [10] **M. Mohammadi Gohari** and M. Amyari. *The Operator-Valued Parallelism and Norm-Parallelism in Matrices*. Indian Journal of Pure and Applied Mathematics, 51(4):1305–1316, 2020. [DOI]
- [11] **M. Mohammadi Gohari** and M. Amyari. *Parallelism in Hilbert  $K(H)$ -Modules*. Tbilisi Mathematical Journal, 12(2):67–75, 2019. [DOI] [arXiv]
- [12] **M. Mohammadi Gohari** and M. Amyari. *Refinements of the Heinz Inequalities for Operators and Matrices*. Mathematica Slovaca, 68(6):1431–1438, 2018. [DOI]
- [13] M. H. Jahandar, M. Elahi Torshizi, H. Farhangfar, and **M. Mohammadi Gohari**. *Using Mathematical Models for the Lactation Curve of Milk and Fat Production in Iranian Holstein Cows*. Indian Journal of Animal Sciences, 82(9):1016–1021, 2012. [DOI]

### Manuscripts in preparation

- [14] **M. Mohammadigohari** and N. Mücke. *Spectral Bounds for Brownian Kernel Ladders: Covering, Localization, and Fixed-Design Minimax Risk*. In preparation, 2026.
- [15] **M. Mohammadigohari** and N. Mücke. *Threshold Geometry and Margin Certificates for Brownian Kernel Ladders*. In preparation, 2026.

## Research Grants and Funded Projects

### Post-Lauream Research Grant

Dec 2025–Jun 2026

Free University of Bozen–Bolzano, Italy

Project: *AI Engineering Suite to Support Agile and Efficient Software Engineering (AI4SWEng)*. Principal Investigator: Prof. Giuseppe Di Fatta. University Research Grant, CUP I53C250000000006.

### Learning Fair Latent Representations with Multi-Task Deep Learning

Nov–Dec 2025

MTDL Start-Up Fund

Short-term funded project on fairness-aware multi-task representation learning and its theoretical and empirical evaluation.

## Technical Expertise

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|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Methods</b>   | RKHSs and variation spaces; functional analysis and operator theory; Rademacher/Gaussian complexity; convex analysis; gradient flows; approximation theory; semidefinite optimization. |
| <b>ML theory</b> | Statistical learning theory; kernel methods; multi-task and representation learning; deep-learning theory; finite and mean-field models.                                               |
| <b>Software</b>  | Python (PyTorch, NumPy, SciPy, scikit-learn, CVXPY), MATLAB, R, $\text{\LaTeX}$ , and Git; reproducible theorem-aligned experiments.                                                   |
| <b>Languages</b> | Persian (native); English (proficient).                                                                                                                                                |